Engineering Ethics Case Studies

- Case study 1: Volkswagen Emissions Scandal

THE UNIVERSITY OF SYDNEY

Presenter: Group 21 Presentation

Volkswagen Emissions Scandal Case Overview

- ❑ VW cheated on test results of emissions.
- ❑ Cars that run on diesel had defeat devices.
- ❑ Deceived regulators and customers all over the world.

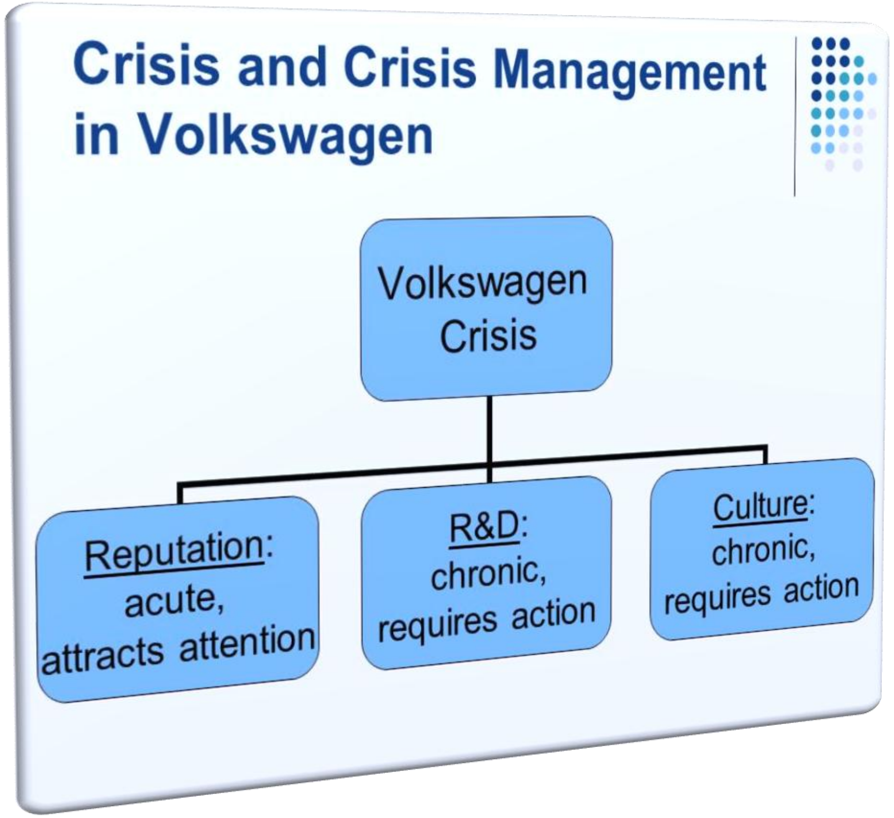
❑Exposure by scandal in 2015 investigations.
❑They cause serious legal and reputational risks (Kaçiu, 2024).

- Malicious misuse of software design.
- Breaking professional ethics.
- Put their profits before the environment.
- Weakened systems of regulatory compliance.
- Organizational-wide systematic malpractice (Fey and Amis, 2023).

- Deontological ethics breached the honesty duty.
- Utilitarian ethics disregarded the effects of environmental damage.
- Integrity and responsibility was compromised by virtue ethics.
- Violation of professional codes of ethics.
- Multi-framework failure apparent (Tanner & Groos, 2022)

Deception Mechanism

- Testing conditions were detected using software.
- Emissions modified under regulatory tests.
- Emission levels were greater when driving normally.
- Faked a false compliance appearance.
- Complicated engineering manipulation was involved (Tanner and Groos, 2022).



- Dynamics to achieve performance objectives.
- Raising tolerance to unethical decision escalation.
- Absence of internal accountability systems.
- Whistleblowing was discouraged by hierarchy.
- The continuation of misconduct was made possible through culture (Tanner & Mansell, 2025).

Regulatory Failure

- There is weak software manipulation detection.
- Dependence on lab techniques of testing.
- Lack of real-life monitoring of emissions.
- Late response to regulation in the world.
- Essential areas of oversight fallacies (Kaçiu, 2024).

- Billions paid out their fine and settlements.
- Executives were charged with criminal cases.
- Distrust among consumers everywhere.
- Extensive damage of brand reputation.
- Expenses on the long run (Fey & Amis, 2023).

Impact on the Engineering Profession

- Erosion of public trust in engineers
- Raised concerns about ethical accountability
- Highlighted need for ethical training
- Questioned integrity of engineering systems
- Professional standards scrutiny increased (Tanner & Groos, 2022)

Ethical Analysis

- Breaking of honesty and transparency.
- Failure to observe environmental responsibility obligations.
- Failure to observe environmental responsibility obligations.
- Group failure instead of individual failure.
- Deterioration of unethical performance (Tanner and Groos, 2022)

Prevention Strategies

Strengthen	Strengthen corporate ethical culture
Enhance	Enhance whistleblower protection systems
Implement	Implement independent compliance audits
Improve	Enhance the engineering process transparency.
Embed	Embed ethics in design decisions (Tanner & Mansell, 2025)

Lessons Learned

- Ethics must guide engineering decisions
- Short-term gains risk long-term losses
- Accountability systems are essential
- Transparency prevents systemic misconduct
- Ethical leadership is critical (Kaçiu, 2024)

Engineering Ethics Case Studies

- Case study 2: The Collapse of the Morandi Bridge in Italy Case

The Collapse of the Morandi Bridge in Italy Case

- Second case examines infrastructure ethics
- Focus on safety and maintenance failures
- Civil engineering responsibilities emphasized
- Public safety consequences severe
- Morandi Bridge collapse analysis follows (Orgnoni et al., 2022)

Morandi Bridge Overview

Bridge collapsed in Genoa Italy 2018

Dozens killed and infrastructure destroyed

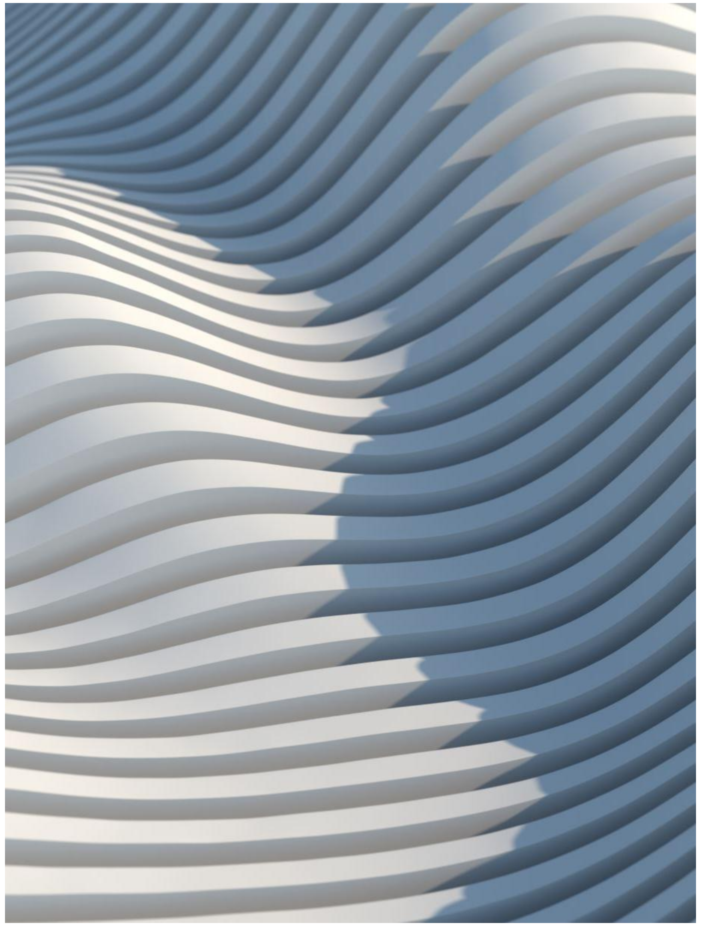
Critical structural failure occurred

Bridge had known maintenance issues

Tragedy highlighted systemic problems (Orgnoni et al., 2022)

Engineering Responsibilities

Ensure	Ensure structural safety and reliability
Conduct	Conduct regular inspections and monitoring
Address	Address deterioration proactively
Maintain	Maintain accurate risk assessments
Protect	Protect public safety priority (D’Angelo et al., 2025)



Ethical Issues

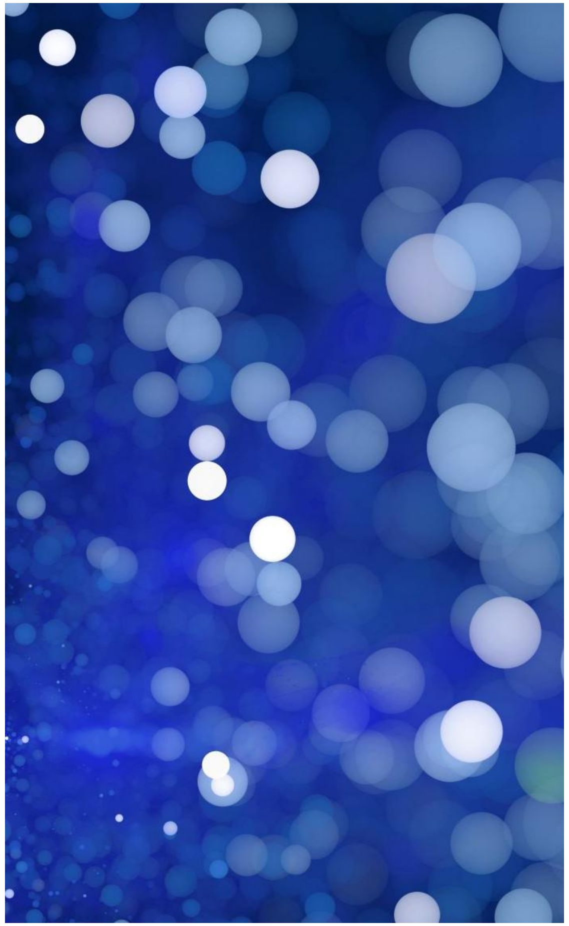
Neglect of maintenance responsibilities

Failure to act on warning signs

Possible cost-cutting decisions

Inadequate monitoring systems

Compromised safety standards (Lavezzo, 2022)



Technical Factors

- Aging infrastructure and material degradation
- Complex cable-stayed design challenges
- Insufficient reinforcement over time
- Loading stresses underestimated
- Design vulnerabilities identified (Orgnoni et al., 2022)

Organizational Failures

- Fragmented responsibility among stakeholders
- Lack of clear accountability structures
- Inadequate communication systems
- Delayed maintenance interventions
- Systemic governance weaknesses (D'Angelo et al., 2025)

Regulatory Context

- Insufficient infrastructure oversight policies
- Weak enforcement of maintenance standards
- Reactive rather than proactive regulations
- Post-collapse legal reforms introduced
- Policy gaps exposed (Lavezzo, 2022)

Consequences

- Death of humans and injuries.
- Disruption of economy and damage of infrastructure.
- Social mistrust towards the government.
- Prosecutions of the responsible parties.
- Trauma of the community over a long period (Migliorini et al., 2025).

Ethical Analysis Bridge

- Violation of duty of care principle
- Neglect of precautionary safety measures
- Failure in risk communication
- Institutional ethical breakdown
- Preventable disaster identified (Orgnoni et al., 2022)

Prevention Strategies Bridge

01

Implement continuous structural monitoring systems

02

Increase funding for maintenance programs

03

Strengthen regulatory enforcement mechanisms

04

Clarify accountability responsibilities

05

Promote safety-first culture (D’Angelo et al., 2025)

Lessons Bridge Case

The infrastructure needs constant monitoring.

Ethics integral to engineering decisions

Neglect leads to catastrophic outcomes

Transparency improves public trust

Preventive action essential (Migliorini et al., 2025)

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Sustainability case study

Sustainability Case 1: Sydney metro

Group21

Source: Sydney Metro Sustainability Report 2022

Introduction

01

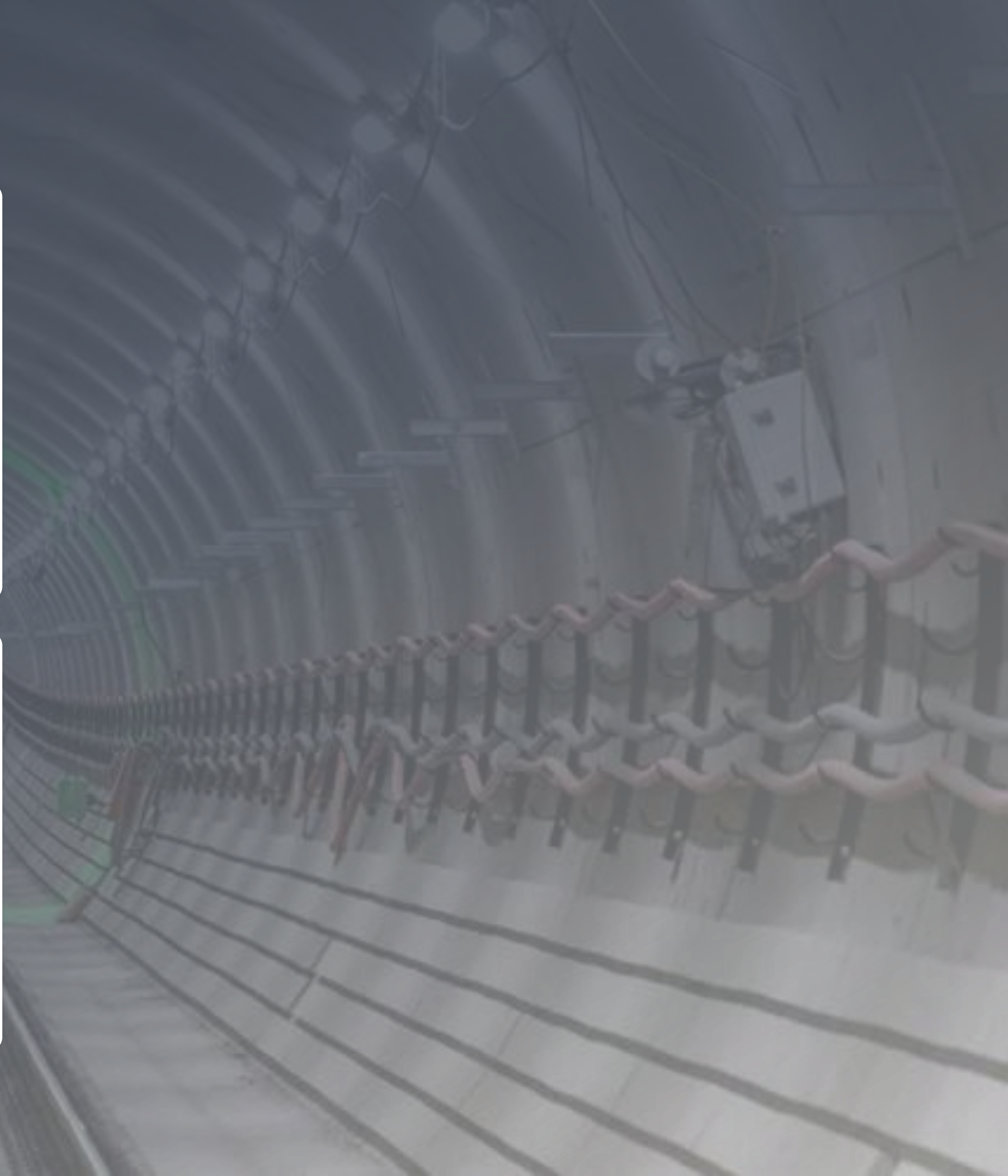
What is Sydney Metro?

Sydney Metro is a major urban rail project in Sydney and a useful case for analysing infrastructure sustainability.

02

How to analyze Sydney metro?

This case study focuses on its materials, resource efficiency, energy performance, and social impacts.



Source: Sydney Metro Sustainability Report 2022

Overview

Outline

01

Materials and Recycle

Reducing virgin resource demand through innovative material substitution and circular economy principles

02

Electricity and Water

Resource efficiency beyond carbon with measurable operational performance targets

03

Social and Economy

Workforce inclusion, training value, and economic participation during delivery

04

NCC 2022 & Conclusion

Beyond minimum compliance and key lessons for future engineers

Materials

Reducing Virgin Resource Demand

Innovative material substitution strategies for lower embodied carbon

43% **Portland Cement**

Replaced with supplementary cementitious materials

72,000+ **Tonnes**

Structural and reinforcing steel deployed across project

90%+ **Australian**

Steel sourced from domestic manufacturers

60%+ **Energy-Reducing**

Structural steel from low-emission processes

1,000+ **Tonnes Recycled Glass**

Used as sand replacement, reducing imported material dependence

Key Takeaways

Lower Embodied Carbon

significant reduction in construction carbon footprint

Reduced Virgin Material Dependence

Circular material flows integrated into procurement

Recycle & Circularity

Reducing Waste and Transport Emissions

Near-complete waste diversion and spoil beneficial reuse

97%

Construction & Demolition Waste Recycled

Industry-leading diversion rate from landfill

~171,865 t

Diverted from landfill

100%

Reusable spoil beneficially reused

985,670 t

Spoil reused (2020–22)

523,380 t

Reused on site

~32,000 Truck Movements Avoided

Significant transport emission reduction

Key Takeaways

- Less landfill burden
- Less virgin extraction
- Less transport demand

Electricity & Carbon

Strongest Decarbonisation Evidence

Operational and construction carbon reduction achievements

100% Zero-Emission Electricity

Metro North West Line operations

133,000+ t

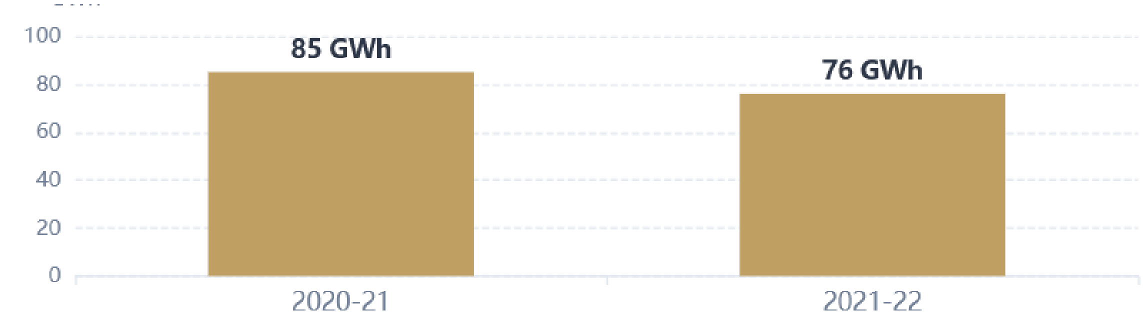
CO_2 emissions saved (2020–21 & 2021–22)

27%

Below BAU

City & Southwest construction emissions reduction

Electricity Consumption



Water Efficiency

Resource Efficiency Beyond Carbon

Comprehensive water management and non-potable source utilization

17.7 ML

Construction water from non-potable sources (2020–22)

Barangaroo Station

35%

Predicted reduction in operational water use

City & Southwest

35%

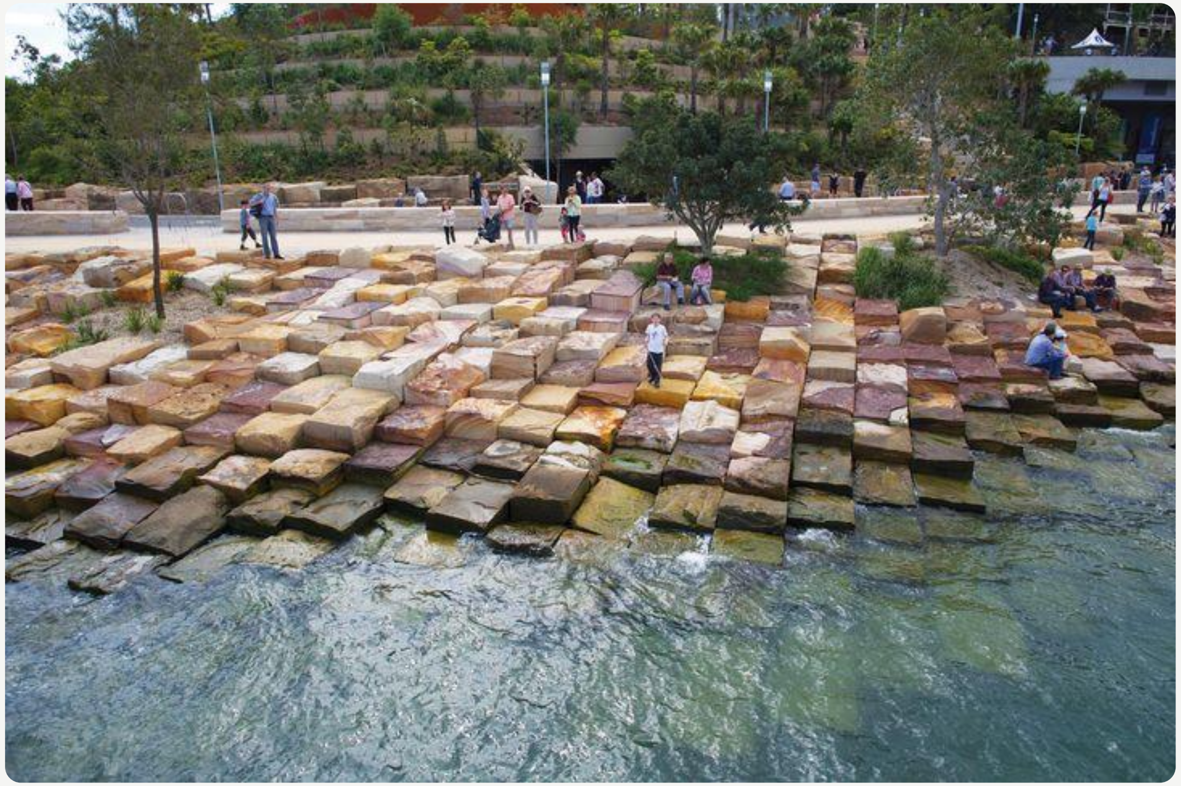
Predicted reduction in potable water demand

Non-Potable Water Use

~14%

Rainwater Tank Capacity

55,000+ L



Key Takeaways

- **Lower potable water demand** through alternative source integration
- **More reuse and substitution** embedded in station design

Social Sustainability

Who Benefits During Delivery?

Workforce inclusion and community value creation

35,526

Workers across all projects

943

Apprentices

175

Trainees

1,027

Aboriginal People

1,484 Long-term Unemployed

Employment pathway programs

46

Community Benefit Initiatives

Key Takeaways

Workforce Inclusion

Diverse employment pathways created

Training & Community Value

Skills development and social investment

Economic Sustainability

Inclusive Procurement and Delivery Value

Economic participation through local and Aboriginal business engagement

1,210

SMEs Engaged

159

Aboriginal Businesses

$105m+

Total Investment

- Aboriginal business engagement
- Workforce training programs
- Employment pathway initiatives

Sustainability Targets

90% achieved

Key Takeaways

Economic Value Through Inclusion

Local participation in supply chain

Delivery Capability

Strong performance on sustainability metrics

NCC 2022

Beyond Minimum Compliance

Performance-led design exceeding regulatory requirements

15% Improvement over NCC Section J

City & Southwest stations and stabling buildings performance achievement

Performance-Led Design

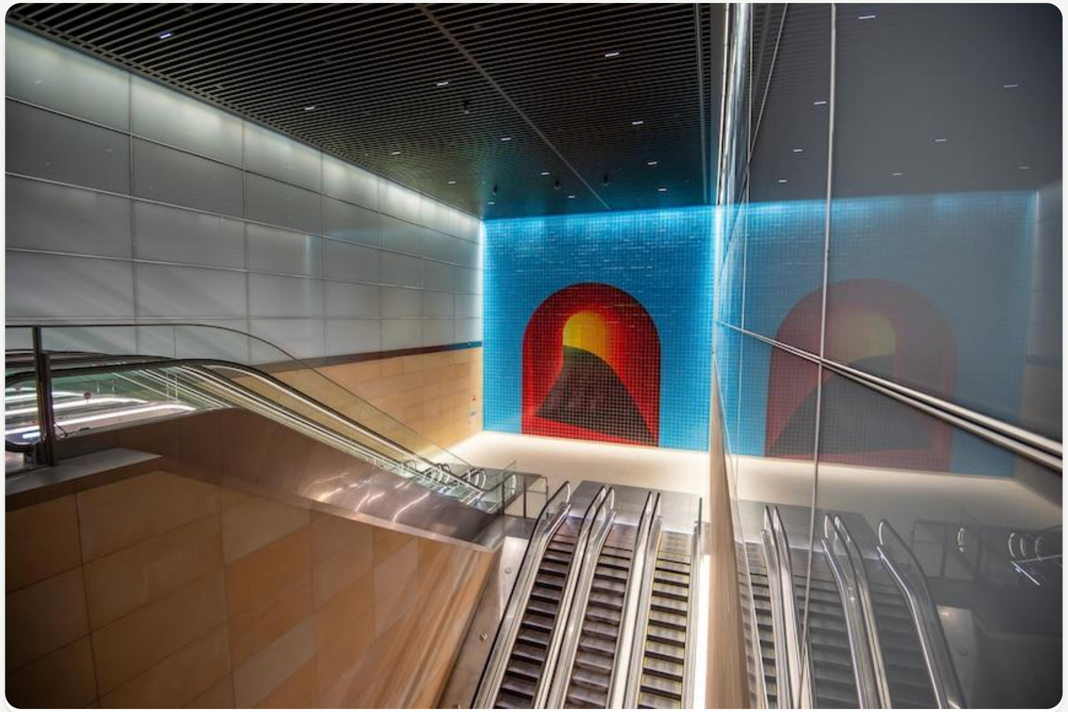
Outcome-focused engineering approach rather than prescriptive compliance

Operational Efficiency

Clear targets for ongoing building performance and energy use

Beyond minimum code compliance

Demonstrated pathway from regulatory minimum to measurable performance outcomes



Key Takeaways

- **From minimum compliance** to aspirational performance targets
- **To measurable performance** with verified outcomes

Conclusion

What Future Engineers Should Learn

Key insights from Sydney Metro's sustainability journey

Strongest Evidence

- Materials and circularity
- Carbon reduction metrics
- Delivery-phase social impact

Main Limitation

Measured outcomes, targets, and package-level results are mixed.

Not all sustainability claims are equally verified.

Main Lesson

Sustainability must be project-wide and measurable. Quantified outcomes across the lifecycle demonstrate true commitment.

> Sydney Metro is a strong sustainability case because it combines measurable resource, carbon and social outcomes, but its evidence is uneven across packages and lifecycle stages.

Sustainability case study

Sustainability Case 2: Barangaroo

Group 21

Sources: Barangaroo Reserve (official); Infrastructure NSW, *Barangaroo: Realising the Vision* (2020)

Introduction:

01 What is Barangaroo?

Barangaroo is a large waterfront redevelopment project in Sydney and a major example of sustainable urban development.

03 How to analyze Barangaroo?

This case study focuses on its environmental systems, public space, and urban liveability outcomes.

Sources: Barangaroo Reserve (official); Infrastructure NSW, *Barangaroo: Realising the Vision* (2020)

Overview

Outline

01

Environmental Systems & Public Space

Carbon-neutral precinct certification, renewable energy, and landscape as sustainability infrastructure

02

Urban Liveability & Economic Value

Connectivity, walkability, jobs, housing, and city productivity outcomes

03

Critical Thinking & NCC 2022

Why Barangaroo is strong but not perfect, and implications for building code compliance

04

Conclusion & Future Lessons

Key takeaways for future engineers on precinct-scale sustainability

Environmental Systems

Carbon, Energy, and Water at Precinct Scale

Integrated sustainability infrastructure across the entire precinct

Climate Active Certified

Carbon neutral precinct

100%

Renewable electricity across precinct

Rooftop Solar

Powering public areas and recycled-water treatment plant

District Cooling

Harbour heat rejection for efficient cooling

Wastewater Treatment & Reuse

Recycled-water systems reducing potable water demand across the precinct

Source: Barangaroo official Sustainability page

Key Takeaways

- **Lower operational carbon impact** through renewable energy
- **Lower potable water demand** via recycling systems
- **Sustainability embedded** in precinct infrastructure

Public Space & Ecology

Landscape as Sustainability Infrastructure

Ecological restoration and public realm transformation

50%+

Public Open Space

6 hectares

Barangaroo Reserve

Restored headland park

75,000+

Native Trees & Shrubs

Indigenous vegetation

Waterfront Restoration

Former industrial waterfront transformed into accessible public landscape

Ecological Restoration

Habitat creation and biodiversity enhancement

Sources: Barangaroo official Public Spaces page; Barangaroo Reserve page

Key Takeaways

- Ecological restoration
- Public access
- Urban amenity improvement

Urban Liveability

Access, Walkability, and Daily Use

Connectivity and public realm design for everyday usability

Waterfront Access

Reconnecting Sydney CBD to the harbour

Wulugul Walk

Public foreshore promenade route

Wynyard Walk

Pedestrian connection to CBD

Multi-Modal Connectivity

Ferry Train Metro

Mixed-Use Precinct Form

24/7 activity beyond office hours

Key Takeaways

- **Better connectivity** to transport and CBD
- **Better walkability** with dedicated pedestrian routes
- **More everyday usability** beyond office hours

Sources: Barangaroo official Accessibility page; Getting to Barangaroo Reserve; Public Spaces page

Economic Value

Jobs, Housing, and City Productivity

Long-term urban value and economic projections

20,000+

Jobs Projected

Once precinct is fully complete

~3,500

Residents Projected

Mixed-use residential population

$2 Billion

Annual Economic Contribution Projected

24/7 Mixed-Use Precinct

Continuous activity generating sustained economic value

Key Takeaways

- ✓ Long-term urban value creation
- ✓ Employment and activity generation

Source: Infrastructure NSW, *Barangaroo: Realising the Vision* (2020)

Critical Thinking

Why Barangaroo Is Strong but Not Perfect

Balanced assessment of achievements and limitations

Strengths

- **Precinct-Scale Systems**
  Not just individual green buildings
- **Integrated Approach**
  Carbon, water, landscape, mobility combined

Limitations

- **Carbon Neutral ≠ Zero Emissions**
  Relies on offsets and accounting
- **Projected Outcomes**
  Economic figures not yet realised
- **Social Inclusion Questions**
  High amenity ≠ automatic equity
- **Replicability Concerns**
  Premium location may limit transferability

Key Takeaways

- **Strong case** with measurable outcomes
- **Strong branding** as sustainable precinct
- **Still needs** critical interpretation

Sources: Barangaroo official Sustainability page

NCC 2022

Beyond Minimum Building Compliance

Performance-led sustainability thinking at precinct scale

Performance-Led Thinking

Beyond minimum compliance logic

A stronger focus on actual performance, not just meeting the minimum rules.

Water Efficiency

Alternative water systems and efficiency targets

Integrated Services

Precinct-scale shared infrastructure

Precinct-Scale Engineering

Systems thinking beyond individual building compliance to precinct-wide performance optimization

Important Note

Directional implication only — not a full NCC compliance audit

Conclusion

What Future Engineers Should Learn

Key insights from Barangaroo's precinct-scale sustainability approach

Strongest Evidence

- Precinct-scale system
- Public space and ecology
- Urban liveability outcomes

Main Limitation

Some results still depend on offsets, some benefits are still projections, and there are still questions about how inclusive this kind of high-end development really is.

Main Lesson

- Design energy, water, space, and transport together
- Focus on real urban benefits, not only technical targets
- Stay critical about trade-offs and limits

> Barangaroo is a strong urban sustainability case at precinct scale, but its carbon-neutral claims and liveability outcomes still need critical interpretation.